. (4) .

x ,

$$B_z = \frac{\mu_0}{2\pi}\frac{p_m}{(z-x)^3}. \tag{11}$$

(10)

$$E = -\frac{r}{2}\frac{dB_z}{dt} = -\frac{r}{2}\cdot\frac{3\mu_0}{2\pi}\frac{p_m}{(z-x)^4}\frac{dx}{dt} \approx -\frac{3\mu_0}{4\pi} r \frac{p_m}{z^4} v \tag{12}$$

$x << z$.

$$j = \frac{1}{\rho}E. \tag{13}$$

$dI = jhdr$,

$$dp'_m = \pi r^2 dI = \pi r^2 j dr \cdot h = -\pi r^2 dr \cdot h \frac{3\mu_0}{4\pi\rho} r \frac{p_m}{z^4} v = -\frac{3\mu_0}{4\rho}\frac{p_m}{z^4} hv r^3 dr. \tag{14}$$

$$p'_m = -\frac{3\mu_0}{4\rho}\frac{p_m}{z^4} hv \int_0^R r^3 dr = -\frac{3\mu_0}{16\rho}\frac{p_m}{z^4} hR^4 v \tag{15}$$

(8) .

(4) .

$$F = -p_m \frac{dB_z}{dz} = -p_m \frac{d}{dz}\left(\frac{\mu_0}{2\pi}\frac{p'_m}{z^3}\right) = \frac{3\mu_0}{2\pi}\frac{p_m p'_m}{z^4} = -\frac{3\mu_0}{2\pi}\frac{p_m}{z^4}\frac{3\mu_0}{16\rho}\frac{p_m}{z^4} hR^4 v = -\frac{9}{32}\frac{\mu_0^2 p_m^2}{\pi\rho z^8} hR^4 v. \tag{16}$$

$$mx'' = -kx - bx' \tag{17}$$

$b = \frac{9}{32}\frac{\mu_0^2 p_m^2}{\pi\rho z^8} hR^4$ (16) .

1.2.3

$$\omega = \sqrt{\omega_0^2 - \beta^2} = \omega_0\left(1 - \frac{\beta^2}{\omega_0^2}\right)^{\frac{1}{2}} \approx \omega_0\left(1 - \frac{\beta^2}{2\omega_0^2}\right) \tag{18}$$

$\beta = b/2m$.

$$\frac{\Delta\omega}{\omega_0} = -\frac{\beta^2}{2\omega_0^2} = -\frac{1}{2km}\left(\frac{9}{64}\frac{\mu_0^2 p_m^2}{\pi\rho z^8} hR^4\right)^2$$

1.2.4

$$\tau = \frac{1}{\beta} = \frac{2m}{b} = \frac{64 m\pi\rho z^8}{9\mu_0^2 p_m^2 hR^4}$$

1.2.5

$$P = \int \rho j^2 \, dV = \int_0^R \rho\left(\frac{3\mu_0 p_m v}{4\pi z^8 \rho} r\right)^2 h \, 2\pi r \, dr = \frac{9h\mu_0^2 p_m^2 R^4}{32\pi z^8 \rho} v^2$$

(16) .